Jorge Alfredo Robles Calderón

Physicist — experimental materials, superconductivity and optics
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SUMMARY

Experimental physicist with independent experience in scientific software. Synthesis and characterization of spinel ferrites; a three-dimensional two-band superconductor solver designed, written and validated from scratch; first-principles nonlinear optics of Weyl semimetals and a computational lensmeter; web and data applications deployed and in use. Physicist, B.Sc. Physics, Universidad Nacional de Colombia (2026).

Areas: condensed matter physics — superconductivity, functional materials, optics. Open to funded master's or PhD positions, and to research or industry roles in scientific software, data and materials R&D.

RESEARCH

Three-dimensional multiband TDGL solver — independent research, UNAL

Design, implementation and validation of a 3D two-band time-dependent Ginzburg–Landau solver for mesoscopic superconductors under field and transport current: gauge-invariant link variables, self-consistent vector potential, self-field current injection, Josephson voltmeter. Octave reference v13.3; Julia solver bit for bit against it; about 56 000 test assertions; lowest Landau level within 1.6 % and H_{c3}/H_{c2} within 0.05 %. Open source (MIT): **juliaTDGL**, two-band release 1.2 and N-band release 2.0 (Kramer–Watts-Tobin relaxation per band, finite interband quasiparticle exchange, anisotropic bands, thermal noise; GPU port, 4–8× faster on large meshes), github.com/georgeotjg/juliaTDGL; documentation at georgeotjg.github.io/juliaTDGL.

Silent failure modes in three-dimensional two-band time-dependent Ginzburg–Landau simulations of geometric superconducting diodes — J. A. Robles Calderón, preprint, Zenodo (2026), doi:10.5281/zenodo.23163207. Eighteen failure modes, each yielding a converged but wrong quantity, each with a test; symmetry-exact diode control; diode efficiency bracketed from −47 % to −21 % [−33, −7] across padding, no converged value claimed. Second paper, manuscript under review: *Rectification in a notched two-band box at low field: two screening regimes and the terms that leave it unchanged*. Two further manuscripts in preparation.

$Zn_{1-x}Mg_xFe_2O_4$ spinel ferrites — bachelor's thesis (trabajo de grado), UNAL, 2026 (152 pp)

Advisor: D. A. Landínez Téllez. Synthesis of the series by glycine-assisted combustion; XRD with Rietveld refinement, SEM/EDX, UV-Vis diffuse reflectance. Analysis automated in Python, including band-gap fits whose window is selected by robustness (scikit-learn). Single-phase $Fd\bar{3}m$ solid solution, coherent domains of 20–26 nm, direct optical gap 1.88–1.91 eV nearly independent of composition. GGA+U band structures (treatment proposed and analysed in the thesis, computed within the research group) place the gap in the Fe–O subsystem.

Nonlinear optics of Weyl semimetals — 2026

First-principles photocurrents in Weyl semimetals: Octopus (real-time TDDFT) and Quantum ESPRESSO + Wannier90 + Julia. Real-time lattice model reproducing the quantized circular photogalvanic response; Octopus validated on silicon including a symmetry null test; GaAs, TaAs and LaAlSi reproduced against the literature with own Julia tools (shift current, 24 Weyl nodes, optical conductivity).

Computational lensmeter

Measurement of spectacle lenses (sphere, cylinder, axis) by transmission deflectometry with a phone camera's RAW sensor and a laptop screen; elliptic inverse solver and thick-lens model. A virtual optical bench simulates the photographs end to end and feeds the same inverse solver; in simulation the prescription is recovered within ±0.12 D.

APPLIED PROJECTS

Geospatial field-data platform. GPS-tagged forms, maps and spatial queries; Next.js, Supabase (PostgreSQL/PostGIS), Leaflet, PWA with offline queue. Deployed and used in the field for interviews, surveys and territorial records.

OCR + local language model. National Astronomical Observatory, UNAL: metadata from scanned documents, with validation that rejects malformed extractions.

Shape editor and run manager for the superconductor solver (JavaScript, Python): 3D sample shapes with live symmetry checks; terminal menu to launch, follow and resume long runs.

Commercial catalogue site. Next.js, TypeScript and Tailwind on Vercel, quotations over WhatsApp. tecnimotos-

villamil.vercel.app

EXPERIENCE

Administrative and Technical Assistant — National Astronomical Observatory, Universidad Nacional de Colombia · 2023–2026
Extension and Technology-Transfer Assistant — Department of Physics, Universidad Nacional de Colombia · 2024–2025
Academic and Scientific Support — Universidad Nacional de Colombia · 2022–2025
Customer Operations — Iké Asistencia · 2023

EDUCATION

B.Sc. Physics — Universidad Nacional de Colombia, Bogotá · 2019–2026. Bachelor's thesis graded 5.0/5.0; GPA 4.0/5.0.
Technical degree, Administrative Assistance — SENA · 2019

SKILLS

Computation: Python (NumPy, SciPy, pandas, scikit-learn, PyTorch, Matplotlib), Julia, MATLAB/Octave, SQL, TypeScript, LaTeX, Git, Linux.
Numerical methods: finite differences on staggered grids, gauge-invariant discretisation, IMEX schemes, Poisson/elliptic solves, preconditioned Krylov solvers, regression and acceptance testing.
First principles: DFT/DFT+U concepts and band-structure analysis; Quantum ESPRESSO and Octopus (real-time TDDFT), in current use.
Laboratory: combustion synthesis, XRD with Rietveld refinement, SEM/EDX, UV-Vis diffuse reflectance.
Data and cloud: data analysis and fitting pipelines in Python; SQL (PostgreSQL), AWS, Docker.
GIS: PostGIS, spatial queries, Leaflet, GPS field data.
Web: Next.js/TypeScript, Supabase (deployed applications).
Languages: Spanish (native), English (B2).